Kmer graph practical

Context

This practical combines assembly and graph visualization: assemble one sample, compare results from different read types, and investigate differences from a related reference.

Learn to use unfamiliar tools: read the README, run the tool without arguments, look on the internet or the associated GitHub repository, or ask your friend Chad or Claude to understand its inputs, options and outputs.

Disclaimer: The reads and genomes are **fully simulated**. The genomes are approximately **500 kb**, so the tools run quickly enough to compare several approaches during the session.

Main step

1. Data examination
2. Assembly
3. Evaluation
4. Graph exploration and comparison with the reference
5. Success!

Choose your reads, tools and parameters freely. You do not need to try everything: investigate results that raise questions, and change approach if you get stuck.

Use the cheatsheet cautiously: check commands or consult it when blocked. The aim is to learn, not copy commands.

Provided PDFs

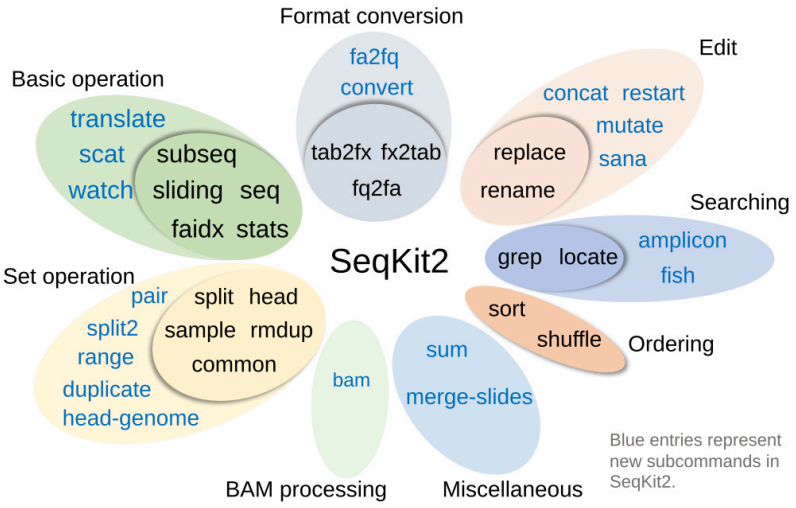
`practical.pdf`	This document: questions and activities.
`cheatsheet.pdf`	Command examples and explanations.
`install_party.pdf`	Tool installation instructions if you cannot use the course Docker image.

Datasets

The data are in `course_data_2026/Kmers_in_a_Graph/`. All three read datasets come from the **same sample**.

`reads/sample_illumina_R1.fastq.gz` `reads/sample_illumina_R2.fastq.gz`	**Illumina like reads:** short paired end reads, with matching mates in the two files.
`reads/sample_ont.fastq.gz`	**Nanopore like reads:** very long reads, with a higher error rate than the HiFi like reads.
`reads/sample_hifi.fastq.gz`	**HiFi like reads:** accurate long reads, shorter than the Nanopore like reads.
`references/reference.fa.gz`	**Related reference genome:** for QUAST and graph comparison. It differs from the sample genome.

Analyze your data



SeqKit2 overview, Wei Shen

Use **SeqKit** to investigate the properties of the datasets. For each dataset, consider the following questions:

- How many bases and reads are there?
- What are the mean read length and the N50?
- What are the lengths of the longest and shortest reads?
- Can you estimate the depth, given that the genome size is approximately 500 kb?

Your first assembly



The goal of this section is to perform an initial assembly using one of the three types of reads provided.

You may choose any of the available assemblers below. The purpose of the practical is to encourage you to take the initiative and make a choice. For this first attempt, if the assembler requires parameters, try to estimate reasonable values without overthinking. At this stage, the quality of the assembly is not critical.

Available assemblers

Short read assemblers

Input: Illumina like paired end reads.

- **SPAdes** assembles short reads using de Bruijn graphs and can use several kmer sizes. You can choose a single kmer size and skip read correction for a faster comparison on these synthetic reads.
- **MEGAHIT** is a fast short read assembler that also uses several kmer sizes. It provides an alternative to compare with SPAdes.
- **GGCAT** constructs compacted de Bruijn graphs and outputs unitigs and links. Unlike assemblers that simplify the graph, it does not remove tips or collapse bubbles. Compare the graph and unitigs at different kmer sizes.

Long read assemblers

Input: Nanopore like reads and/or PacBio HiFi like reads, according to the tool.

- **Flye** builds a repeat graph from long reads. You can use it separately with the Nanopore like or HiFi like reads.
- **hifiasm** assembles PacBio HiFi reads. It can also use ultralong Nanopore reads alongside HiFi reads to help resolve repeats.
- **Verkko** combines accurate long reads and ultralong Nanopore reads in a hybrid assembly.

Exploration (optional)

Paired end versus unpaired reads

Short read assemblers can use pairing information to improve an assembly. Try ignoring pairing information while keeping the same input reads, and compare the results. Does pairing affect contigs, scaffolds, or both?

Kmer sizes

Assemblers relying on kmers can produce different results depending on the value of k. Experiment with kmer sizes or examine intermediate assemblies. Does a larger k resolve every repeat? How does the graph change as you vary k?

Read length and hybrid assembly

Compare assemblies from the different technologies. Which repeats can long reads resolve that short reads cannot? Can read accuracy compensate for reads that do not span a repeat?

You can also combine the HiFi like and Nanopore like reads with hifiasm or Verkko. Compare the hybrid result with assemblies from each technology alone.

Assembly evaluation

Once you obtain an assembly, assess the quality of its contigs. You can use kmer based evaluation, reference based evaluation and graph exploration.

kmer based evaluation



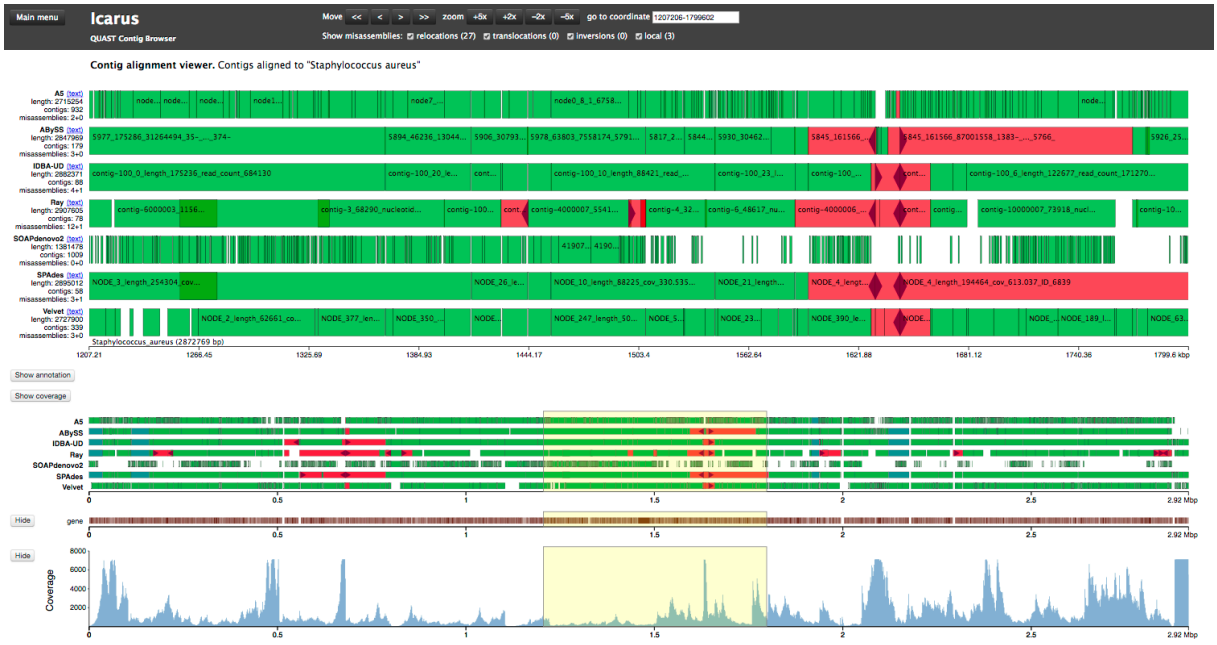
Freddie Mercury, Carl Lender, Creative Commons Attribution ShareAlike 3.0

MerquryFK evaluates an assembly by comparing its kmers with those found in accurate reads from the same sample.

First, use **FastK** to count kmers from the Illumina like reads. Then use **MerquryFK** with the resulting database and your assembly.

Examine the estimated consensus quality, kmer completeness and copy number spectra. Frequent read kmers should generally be represented in the assembly. Rare kmers can reflect sequencing errors.

Reference based evaluation



QUAST / Icarus: official web screenshot (Staphylococcus aureus).

https://quast.sourceforge.net/quast.html

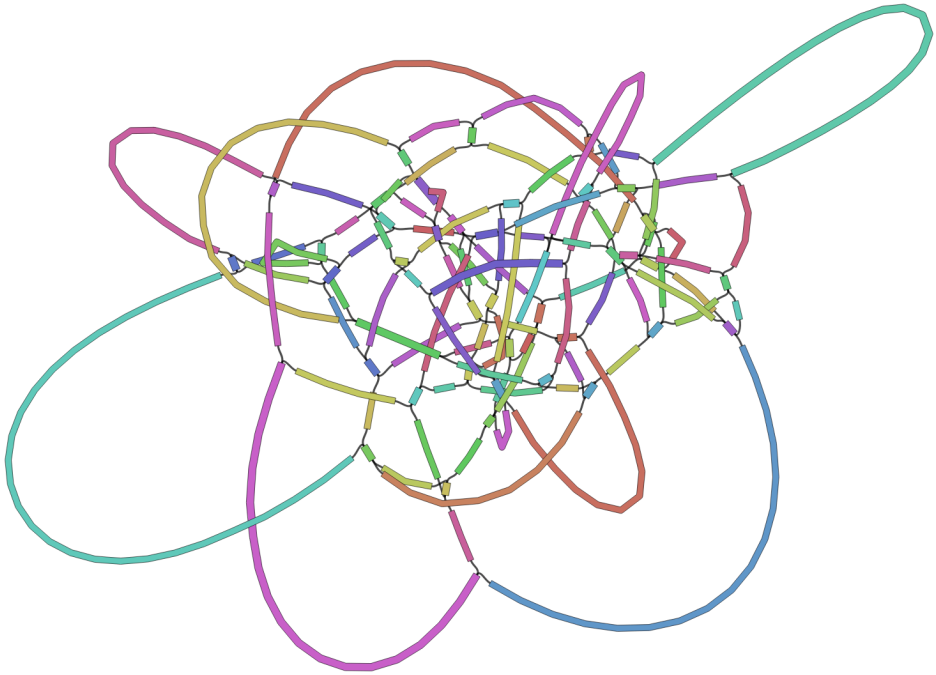
QUAST compares an assembly with a reference genome using sequence alignments. The supplied reference is a **related genome**, not the exact genome of your sample.

Compare several assemblies of your sample in the same analysis. Examine the contig alignment viewer as well as the summary statistics.

- Where does QUAST flag a disagreement with the reference?
- What percentage of the reference is covered by the contigs?
- How contiguous is the assembly? Compare N50 and NGA50.
- How close are the assembled sequences to the reference?
- Does the assembly with the largest N50 also recover the most reference sequence?

Differences from this reference can reflect genuine genomic variation as well as assembly errors. Use read based evaluation and graph exploration to investigate them. A disagreement with the reference is not automatically an assembly error.

Visualizing assembler graphs with Bandage



Bandage opens assembly graphs directly. The course image provides **BandageNG**. Use an assembler's GFA or FASTG graph, such as the SPAdes or Flye assembly graph, to retain its nodes and links.

You can also use **GGCAT** to build a de Bruijn graph from reads or assemblies and export it in **GFA format with graph links** for BandageNG. Try different kmer sizes, or build the union graph of your assembly and the reference.

Load the graph and draw it. Inspect branches, node lengths and depth where available. Select nodes to examine or export their sequences. Compare graphs from different assemblers or kmer sizes.

Visualizing de Bruijn graphs



Vizitig can build and visualize a de Bruijn graph directly from FASTA files. Optionally import a **GGCAT** graph to retain kmer abundance annotations, when available.

For a read graph, examine how connectivity changes with the kmer size. For a graph rebuilt from assembly FASTA files, remember that the graph represents sequence sharing at the chosen k. It does not preserve every decision in the assembler's original graph.

- Is the graph well connected?
- Does it have many branching nodes?
- Which branches could correspond to unresolved repeats?
- How does changing k affect the graph?
- What information would help resolve an ambiguous path?

Graph coloring

Give your assembly FASTA and the reference FASTA directly to **Vizitig** to build their union graph. Keep the two inputs in separate groups to colour shared sequence and sequence found only in the assembly or reference. Alternatively, build the union with **GGCAT** and import it into Vizitig.

Look for small bubbles and larger alternative paths. Inspect their sequences, lengths, connections and orientations.

- Can you find a candidate single nucleotide difference? Compare the sequences of the two paths with Vizitig's aligner.
- Can you find a candidate insertion, deletion or inversion?
- Do the alternative paths connect the same shared flanks?
- Could a difference instead reflect missing reads or an assembly error?
- Does the interpretation remain consistent in an assembly from another technology or a graph built with another kmer size?

Compare and repeat

Try different datasets, combinations of datasets, assemblers and parameters. Evaluate the resulting assemblies with **MerquryFK** and **QUAST**. Explore the graphs with **Bandage** and **Vizitig**.

Once you have compared your assemblies, consider these questions:

1. What are the drawbacks of short read assembly?
2. What are the drawbacks of long read assembly?
3. What are the drawbacks of HiFi read assembly?

How do pairing, read length, accuracy and depth affect your results? What do the graphs explain that assembly statistics alone do not?

Treasure hunt (hard)

If you are brave enough, you can try to assemble by hand this set of "reads":

```
CAGATGTCCT
GACCTTTTCT
TAAGATCTTT
TCTTTAGCCG
TTTCTTCTTT
GAGCTTTACT
TTACTTCATT
TCATTCACAT
CACATGAGCT
GTCCTGAACA
GAGCTGATCA
TCTTTCAGAT
AGCCGGACCT
TATGATCATT
TCATTAGGCG
AGGCGGAGCT
```

HINT: There are NO sequencing errors

HINT: Overlaps are at least 5 bases long